

Software Test Plan (STP)

Project: Online Internship & Placement Preparation Portal

Version: 1.0

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Revision History

Version	Date	Author	Change Summary
1.0	15-09-2026	Vyshali Aytha, Vismaya Harish, Thejus A Reddy, Yashas Sadananda	Initial Software Test Plan draft, derived from SRS v1.0 and Architecture & Design Spec v1.0

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Pavithra P		

1. Introduction

Purpose:

This document defines the Software Test Plan (STP) for the Online Internship & Placement Preparation Portal (OIPP). It outlines the test objectives, scope, strategy, resources, schedule, and responsibilities for verifying that the system meets its functional and non-functional requirements, and is built consistently with its Layered Architecture.

Scope:

Testing covers all eight functional modules of the portal: student registration/authentication, profile & resume management, internship/placement listing search, application submission & tracking, mock tests with auto-scoring, the interview-preparation module (question bank and mock-interview booking), the recruiter dashboard (job posting & shortlisting), and placement-cell admin functions (recruiter approval & reporting). It also covers the associated non-functional requirements (performance, reliability, security, scalability, accessibility) and the security requirements (OIPP-SR-001 to OIPP-SR-005). Excluded from this project's scope, and therefore from this test plan, are: institutional single-sign-on (SSO), automated resume parsing, integrated video-conferencing for mock interviews, payment-based premium test packs, large-scale (thousands of concurrent users) load handling, payroll/HR onboarding of recruiting companies, and any activity after a candidate formally joins an organisation.

References:

- Software Requirements Specification (SRS) — Online Internship & Placement Preparation Portal, dated 09-09-2026

- Software Architecture and Design Specification — Online Internship & Placement Preparation Portal, dated 14-09-2026
- Requirements Traceability Matrix (RTM)
- IEEE 42010 (Architecture Description)
- OWASP Top 10
- WCAG 2.1 (Accessibility) — referenced for OIPP-NF-005

Definitions:

SRS (Software Requirements Specification), UI (User Interface), API (Application Programming Interface), OTP (One-Time Password), JWT (JSON Web Token), CGPA (Cumulative Grade Point Average), RTM (Requirements Traceability Matrix), RBAC (Role-Based Access Control), OIPP (Online Internship & Placement Preparation Portal), STP (Software Test Plan), TC (Test Case), NFR (Non-Functional Requirement).

2. Test Items

The following components, mapped directly to the Business-Layer modules of the portal, are the items under test:

- Web UI Component — responsive, role-based dashboard (Student, Recruiter, Admin views)
- AuthModule — registration/OTP, login/session issuance, account lockout, recruiter account requests
- ProfileModule — student profile fields, resume upload/replace, generated PDF resume
- ListingModule — active listing display and filtering (role, location, CGPA)
- ApplicationModule — application submission, duplicate prevention, status tracker
- AssessmentModule — timed mock tests, auto-scoring, score history
- InterviewPrepModule — interview question bank, mock-interview slot booking, admin feedback
- RecruiterModule — listing creation (post-approval), applicant viewing/shortlisting
- AdminModule — recruiter account approval, placement summary reporting
- Database (Data Layer) — single relational database (e.g. PostgreSQL/MySQL) holding all persistent data
- Email/SMS Gateway integration point — used only for OTP delivery (external service; tested at the integration boundary only, per Section 4 below)

3. Features to be Tested

Features are mapped to the defined requirement IDs and test cases (Functional: OIPP-F-####, Non-Functional: OIPP-NF-####, Security: OIPP-SR-####):

Req ID	Feature / Requirement (short)	Priority
OIPP-F-001	Student self-registration with OTP verification	High
OIPP-F-002	Login with email/password; session token issuance	High
OIPP-F-003	Account lockout after 5 consecutive failed logins	Medium
OIPP-F-020	Recruiter account request (Pending until admin approval)	High
OIPP-F-004	Structured student profile form (academic/skill/project)	High

Req ID	Feature / Requirement (short)	Priority
OIPP-F-005	Resume upload and replace (PDF/DOC)	High
OIPP-F-006	Auto-generated downloadable PDF resume from profile	Medium
OIPP-F-007	Display active internship/placement listings	High
OIPP-F-008	Filter listings by role, location, minimum CGPA	Medium
OIPP-F-010	Apply to a listing; prevent duplicate applications	High
OIPP-F-011	Application status tracker (Applied/Shortlisted/.../Rejected)	High
OIPP-F-012	Timed multiple-choice mock test with auto-scoring	High
OIPP-F-013	Score history across all attempted mock tests	Low
OIPP-F-014	Searchable interview question bank by company/domain	Medium
OIPP-F-015	Mock-interview slot booking + admin text feedback	Low
OIPP-F-016	Recruiter creates/publishes a listing (post-approval)	High
OIPP-F-017	Recruiter views applicants; marks Shortlisted/Rejected	High
OIPP-F-018	Admin approves/rejects pending recruiter account request	High
OIPP-F-019	Admin generates placement summary report (filterable)	Medium
OIPP-NF-001	Page load / search results render $\leq 3s$ (90th percentile)	High
OIPP-NF-002	System available during declared demo/evaluation hours	Medium
OIPP-NF-003	Secure storage: hashed passwords, encrypted connections	High
OIPP-NF-004	Supports at least 100 concurrent users	Medium
OIPP-NF-005	Key student flows usable with screen reader / colour contrast	Medium
OIPP-SR-001	HTTPS (TLS) mandatory for all client-server connections	High
OIPP-SR-002	Role-based access control (Student, Recruiter, Admin)	High
OIPP-SR-003	Passwords stored using salted, industry-standard hashing	High
OIPP-SR-004	Security-relevant events logged with timestamp	Medium
OIPP-SR-005	JWT session tokens with defined expiry; re-login enforced	Medium

4. Features Not to be Tested

The following are outside the scope of this project and this test plan:

- Institutional Single Sign-On (SSO) — not implemented in current scope
- Automated resume parsing — future/optional enhancement, not implemented
- Integrated video-conferencing for mock interviews — future/optional enhancement, not implemented
- Payment-based premium test packs — future/optional enhancement, not implemented
- Large-scale load handling (thousands of concurrent users) — out of scope; only up to 100 concurrent users (OIPP-NF-004) is tested
- Payroll/HR onboarding of recruiting companies and any post-joining activity — out of scope

- Internal implementation of the third-party Email/SMS Gateway — only the integration point (OTP request/response) is tested; the gateway's own delivery infrastructure is assumed to work and is a vendor responsibility
- Native mobile applications — the portal is a responsive web application only; no native iOS/Android app exists to test

5. Test Approach / Strategy

Levels

- Unit tests — module-level, exercising each of the eight Business-Layer modules (AuthModule, ProfileModule, ListingModule, ApplicationModule, AssessmentModule, InterviewPrepModule, RecruiterModule, AdminModule) in isolation
- Integration tests — module-to-module flows through the API layer, following the two key sequence flows: (1) Student login → browse/filter listings → apply (AuthModule ↔ ListingModule ↔ ApplicationModule), and (2) Recruiter account request → Admin approval → listing post (AuthModule ↔ AdminModule ↔ RecruiterModule)
- System tests — end-to-end, role-based workflows across the full stack (Web UI → API → Business Layer → Database) for Student, Recruiter, and Admin
- Acceptance tests — validated directly against each requirement's Acceptance Criteria (AC-OIPP-F-###, AC-OIPP-NF-###, AC-OIPP-SR-###)

Types

- Functional testing — all OIPP-F-### requirements across the eight functional modules
- Regression testing — re-run of the core suites (Auth, Application, Recruiter approval chain) after each defect fix, given the tight dependency chain (e.g. OIPP-F-016 depends on OIPP-F-018, which depends on OIPP-F-020)
- Performance testing — response time (OIPP-NF-001) and concurrent-user load (OIPP-NF-004)
- Usability / Accessibility testing — screen-reader and colour-contrast checks on key student-facing flows (OIPP-NF-005)
- Security testing — see Section 5.1 below

Entry Criteria

- A stable build is deployed to the single application server and connected to the relational database
- Test data is available: dummy student/recruiter/admin accounts, sample resumes, sample listings, and question-bank entries
- The Email/SMS Gateway is configured in sandbox/test mode for OTP delivery
- The test environment (browser matrix, tools) matches the project's defined technology stack

Exit Criteria

- All High-priority functional requirements are implemented and verified
- No critical NFR failures remain open
- The RTM (Section 13 below) shows all planned test cases with Status = P (Pass) or a documented, accepted A (Accepted deviation)

5.1 Security Validation

Security testing covers the STRIDE threat categories and the defined security requirements (OIPP-SR-001–005):

- Spoofing — verify passwords are hashed (not stored/transmitted in plain text) and JWT session validation is enforced on every authenticated request (OIPP-SR-003, OIPP-SR-005)
- Tampering — verify HTTPS (TLS) is enforced on all client–server communication and HTTP requests are redirected to HTTPS (OIPP-SR-001)
- Repudiation — verify security-relevant events (login, logout, role change, recruiter approval) are logged with a timestamp (OIPP-SR-004)
- Information Disclosure — verify a user can only read data permitted by their role, over HTTPS (OIPP-SR-001, OIPP-SR-002)
- Denial of Service — verify account lockout after 5 consecutive failed login attempts as a partial brute-force mitigation (OIPP-F-003); full DoS protection is out of scope at mini-project scale
- Elevation of Privilege — verify role-based access control (RBAC) rejects actions outside a role's permitted endpoints, for all three roles (OIPP-SR-002)
- Session expiry — verify expired JWT tokens are rejected by the API and force re-login (OIPP-SR-005)
- No passwords, tokens, or other sensitive data appear in application logs

6. Test Environment

Hardware

Standard desktop, laptop, and mobile devices with a modern web browser; no specialised hardware is required.

Software

- Web front end: responsive role-based dashboard (Student, Recruiter, Admin views)
- API layer: REST/JSON API (e.g. Node.js/Express or Spring Boot)
- Database: single relational database (e.g. PostgreSQL or MySQL)
- Session management: JWT-based tokens with defined expiry
- Transport: TLS/HTTPS for all client–server traffic
- Email/SMS Gateway: sandbox/test account, used only for OTP delivery

Browser / Device Matrix

Chrome, Firefox, and Edge on both desktop and mobile viewports.

Tools

- Selenium or Playwright — UI automation for role-based dashboard flows
- Postman — API testing of documented endpoints (POST /api/auth/register, POST /api/auth/verify-otp, POST /api/applications, POST /api/interview-slots/{slotId}/book, and other module endpoints)
- JMeter or k6 — performance/load testing (response time target, 100 concurrent users)
- Lighthouse / axe DevTools — accessibility checks against WCAG 2.1 (OIPP-NF-005)
- GitHub Issues (or Jira, if available to the team) — defect tracking, scaled for a small student team

Test Data

- Dummy student accounts with varying CGPA and academic/skill/project profiles
- Dummy recruiter accounts in both Pending and Approved states
- Sample resumes in PDF and DOC formats (including an oversized file to test the upload size limit)
- Sample internship/placement listings with varying role, location, and CGPA-eligibility values
- Sample interview question-bank entries categorised by company/domain
- Sample mock-interview slots (available, booked)
- A pre-loaded question bank for mock tests, with known correct answers for score-verification

7. Test Schedule

Milestone	Date
Test case design	16-Sep-2026 to 20-Sep-2026
Test environment setup	21-Sep-2026
Unit & integration test execution	22-Sep-2026 to 30-Sep-2026
System test execution	01-Oct-2026 to 08-Oct-2026
Security & performance testing	09-Oct-2026 to 12-Oct-2026
Regression & defect-fix cycle	13-Oct-2026 to 16-Oct-2026
Acceptance testing (UAT / evaluator demo)	17-Oct-2026 to 20-Oct-2026
Final Test Summary Report & sign-off	21-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated), keyed to the existing TC-xxx-## references
- Test Scripts (Selenium/Playwright, Postman collections, JMeter/k6 plans)
- Test Data sets
- Test Execution Logs
- Defect Reports
- Updated Requirements Traceability Matrix (RTM), with Status column moved from N to P/A
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Vyshali Aytha	Prepare and maintain the test plan; coordinate test execution and reporting across the team
Test Engineer	Vismaya Harish	Design and execute test cases against the RTM; log and track defects

Role	Name	Responsibility
Developer / Fix Owner	Thejus A Reddy	Support defect triage and fixes; maintain the AuthModule/ApplicationModule build under test
Developer / Build Owner	Yashas Sadananda	Maintain the test environment (app server, database, sandbox gateway); support defect fixes
Course Coordinator	Pavithra P	Approve the test plan and test results; sign off on readiness for evaluation

10. Risks and Mitigation

Risk	Mitigation
Single application server is a single point of failure	Accepted for mini-project scale; schedule test runs to avoid overlap with maintenance windows, and monitor uptime per OIPP-NF-002
Fake or low-quality recruiter/listing postings	Verify the admin-approval gate (OIPP-F-018) is tested and passing before running listing-visibility test cases (OIPP-F-016 depends on OIPP-F-018)
Duplicate or accidental repeat applications	Dedicated duplicate-prevention test case (TC-App-01) run before other ApplicationModule tests
Brute-force login attempts	Dedicated account-lockout test case (TC-Auth-03) verifies lockout after 5 consecutive failures
Unauthorized access to another user's data	Full RBAC test suite (TC-Sec-03) run across all three roles before system testing begins
Small student team with limited time in a single academic term	Prioritise High-priority requirements first; Low-priority items (e.g. OIPP-F-013 score history) may be deferred if the schedule slips
Dependency on the external Email/SMS Gateway for OTP testing	Use a sandbox/test account to avoid real SMS/email costs and reduce flakiness during automated test runs
Delay in stable build delivery	Request early smoke builds from the developer team members before each milestone

11. Assumptions & Dependencies

- A stable build on the single application server and single relational database will be available for each test cycle
- Email/SMS Gateway sandbox credentials will be available before OTP-dependent test execution (OIPP-F-001, OIPP-F-020)

- Dummy student, recruiter, and admin accounts, and sample resumes/listings, will be provided or seeded before execution begins
- No native mobile apps exist to test — only the responsive web application on modern browsers (Chrome, Firefox, Edge)
- The test team has access to the same single-server/single-database environment as the development team, consistent with the project's single-server constraint
- Third-party components (Email/SMS Gateway) are assumed to function correctly at their own service boundary

12. Suspension & Resumption Criteria

Suspend testing if:

- The application server or test environment is unavailable for more than 4 hours
- The build is too unstable, blocking more than 30% of planned test cases
- A critical security flaw is found (e.g. an RBAC bypass allowing cross-role data access) that blocks further meaningful testing

Resume testing if:

- Blocking defects are resolved and re-verified
- The environment is stabilised and the application server/database are reachable

13. Test Case Management & Traceability

The RTM maps every requirement to its owning module and existing test case reference, ensuring full traceability from requirement to test. The Status column will be updated from N (Not tested) to P (Pass) or A (as applicable) as execution proceeds:

Req ID	Requirement (short)	Module	Test Case(s)	Status
OIPP-F-001	Register with OTP	AuthModule	TC-Auth-01	N
OIPP-F-002	Login & session token	AuthModule	TC-Auth-02	N
OIPP-F-003	Lockout after 5 failures	AuthModule	TC-Auth-03	N
OIPP-F-020	Recruiter account request	AuthModule	TC-Auth-04	N
OIPP-F-004	Structured profile form	ProfileModule	TC-Prof-01	N
OIPP-F-005	Resume upload/replace	ProfileModule	TC-Prof-02	N
OIPP-F-006	Generate PDF resume	ProfileModule	TC-Prof-03	N
OIPP-F-007	List active postings	ListingModule	TC-List-01	N
OIPP-F-008	Filter listings	ListingModule	TC-List-02	N
OIPP-F-010	Apply, prevent duplicates	ApplicationModule	TC-App-01	N
OIPP-F-011	Application status tracker	ApplicationModule	TC-App-02	N
OIPP-F-012	Mock test auto-scoring	AssessmentModule	TC-Assess-01	N

Req ID	Requirement (short)	Module	Test Case(s)	Status
OIPP-F-013	Score history view	AssessmentModule	TC-Assess-02	N
OIPP-F-014	Interview question bank	InterviewPrepModule	TC-Interview-01	N
OIPP-F-015	Mock-interview booking	InterviewPrepModule	TC-Interview-02	N
OIPP-F-016	Recruiter posts listing	RecruiterModule	TC-Recruit-01	N
OIPP-F-017	Shortlist applicants	RecruiterModule	TC-Recruit-02	N
OIPP-F-018	Approve recruiter	AdminModule	TC-Admin-01	N
OIPP-F-019	Placement summary report	AdminModule	TC-Admin-02	N
OIPP-NF-001	Response time target	Web UI / API	TC-Perf-01	N
OIPP-NF-002	Availability during evaluation	Infra	Ops log review	N
OIPP-NF-003	Secure data storage	Infra	TC-Sec-01	N
OIPP-NF-004	100 concurrent users	Web UI / API	TC-Perf-02	N
OIPP-NF-005	Accessibility of key flows	Web UI	TC-UX-01	N
OIPP-SR-001	HTTPS enforcement	Infra	TC-Sec-02	N
OIPP-SR-002	Role-based access control	AuthModule	TC-Sec-03	N
OIPP-SR-003	Password hashing	AuthModule	TC-Sec-04	N
OIPP-SR-004	Security event logging	Infra	TC-Sec-05	N
OIPP-SR-005	JWT session expiry	AuthModule	TC-Sec-06	N

14. Test Metrics & Reporting

Metrics collected

- % of test cases executed (against the 29 requirement IDs in the RTM)
- % passed / failed per module
- Defect density per module (AuthModule, ProfileModule, ListingModule, ApplicationModule, AssessmentModule, InterviewPrepModule, RecruiterModule, AdminModule)
- Defect aging (time open until resolved)
- Requirement coverage (RTM Status distribution: N / P / A)
- NFR compliance: 90th-percentile response time against the 3s target (OIPP-NF-001); peak concurrent users sustained against the 100-user target (OIPP-NF-004)

Reports

- Weekly execution status report, circulated to the QA Lead and Course Coordinator
- Final Test Summary Report, submitted alongside the updated RTM with all Status values finalised

15. Approvals

Role	Name	Signature / Date
QA Lead	Vyshali Aytha	
Dev Lead	Thejus A Reddy	
Product Owner	Yashas Sadananda	
Course Coordinator	Pavithra P	